EXAMPLE: CUBIC POLYNOMIAL TRAJECTORY PLANNING

A JOINT OF A 6 DOF ROBOT MANIPULATOR IS REQUIRED TO MOVE FROM AN INITIAL ANGLE OF 20° TO A FINAL ANGLE OF 90° IN 7s. CALCULATE THE JOINT ANGLE AND VELOCITY AT 2, 4 AND 6 SECONDS, USING A THIRD ORDER POLYNOMIAL

<u>SOLUTION:</u> SUBSTITUTE BOUNDARY CONDITIONS INTO THIRD-ORDER POSITION AND RESULTING VELOCITY EQUATIONS. I.E.

$$\theta(t_i) = C_0 = 20^\circ$$

$$\theta(t_f) = C_0 + C_1(7) + C_2(7^2) + C_3(7^3) = 90$$

$$\dot{\theta}(t_i) = C_1 = 0 \quad \text{(ASSUMED INITIAL VELOCITY)}$$

$$\dot{\theta}(t_f) = C_1 + 2C_2(7) + 3C_3(7^2) = 0 \quad \text{(ASSUMED FINAL VELOCITY)}$$

THEREFORE, THE COEFFICIENTS ARE.

$$C_0 = 20$$

$$C_1 = 0$$

$$C_2 = \frac{3(90-20)}{7^2} = 4.29$$

$$C_3 = \frac{-2(90-20)}{7^3} = -0.41$$

THIS GIVES THE FOLLOWING POSITION AND VELOCITY EQUATIONS

$$\theta(t) = 20 + 4.29t^2 - 0.41t^3$$

$$\dot{\theta}(t) = 8.58t - 1.23t^2$$

AT $t = 2$, $\theta(2) = 33.88^\circ$ AND $\dot{\theta}(2) = 12.24^\circ/s$

AT $t = 4$, $\theta(4) = 62.40^\circ$ AND $\dot{\theta}(4) = 14.64^\circ/s$

AT $t = 6$, $\theta(6) = 85.88^\circ$ AND $\dot{\theta}(6) = 7.20^\circ/s$